

Day 10 Plan

Maths: Time, Speed & Distance, Trains, Boats · Reasoning: Mathematical Operations, Missing Numbers · Science: Chemistry 2: Chemical Reactions; Acids, Bases & Salts

Today's schedule (about 5½ hours with breaks)

#	Block	Time	What you do	Where
0	Day 9 recall	15 min	10 questions from Day 9, no notes. Redo yesterday's mistakes.	Page 1-2
1	Maths	75 min	5 simplification questions for speed → formulas & methods (10 min) → 2-3 book questions per pattern, timed → 10 Testbook PYQs on this topic → log mistakes	Section A · book · Testbook
-	Break	10 min	Away from the screen	-
2	Reasoning	45 min	Learn the method (10 min) → 15 timed questions, Testbook PYQs first → log	Section B · Testbook
-	Break	10 min	Away from the screen	-
3	Science	110 min	8 lessons with quick checks (≈65 min) → 20-question practice set → Testbook PYQs: chemical reactions; acids, bases & salts (S tier)	Section C · Testbook
-	Break	10 min	Away from the screen	-
4	General awareness	45 min	20 current-affairs MCQs (20 min) → static GA: Indian geography 2 — national parks, dams, minerals, states — Testbook GA PYQs on it (25 min)	CA PDF · Testbook
5	Close	15 min	Redo today's wrong questions · write 3 rules for tomorrow · send me your scores	Mistake log

Short day? Cut in this order: Testbook science PYQs → reasoning down to 10 questions → static GA down to 10 min. Never cut recall, maths or the science lessons. · Exam pace (average 54 sec a question): GA ~30 sec · Science ~45 · Reasoning ~50 · Maths ~65.

Block 0 — Day 9 recall (answers on the last page)

#	Question	Your answer
1	Name two substances that sublime.	
2	Method to separate cream from milk?	
3	Which mixture shows the Tyndall effect?	
4	Who discovered the neutron?	
5	Maximum electrons in the M shell?	
6	Valency of aluminium?	
7	Formula of calcium hydroxide?	
8	Molecular mass of CaCO_3 ?	
9	Avogadro's number?	
10	Maths: A 10 days, B 15 days — together?	

Fewer than 8 right? Re-read only the lessons you missed — not the whole chapter.

A · MATHS — Time, Speed & Distance, Trains, Boats

Core idea

Every question here is **distance = speed × time**, with one twist per type: trains add their own **length** to the distance, two moving objects use **relative speed**, and boats add or subtract the **stream's speed**. Convert units first: **km/h × 5/18 = m/s**.

Formulas

Concept	Formula / rule
Units	km/h → m/s: $\times 5/18$ · m/s → km/h: $\times 18/5$
Average speed, equal distances	$2xy/(x + y)$ (not $(x + y)/2$)
Relative speed	Opposite directions: add · same direction: subtract
Train passes a pole / man	Time = train length ÷ speed
Train passes a platform / bridge	Time = (train + platform) ÷ speed
Two trains cross	$(L_1 + L_2) \div$ relative speed
Boats	Downstream = $u + v$ · upstream = $u - v$ · boat $u = (D + U)/2$ · stream $v = (D - U)/2$
Late / early	Distance = $(S_1 S_2 \div (S_2 - S_1)) \times$ total time difference
Constant distance	Time $\propto 1/\text{speed}$: speed $\times 3/4 \rightarrow$ time $\times 4/3$

Patterns — the method and one solved example each

Pattern	Method	Solved example
1. Units	$\times 5/18$	54 km/h = $54 \times 5/18 =$ 15 m/s
2. Average speed	$2xy/(x + y)$	Goes at 40, returns at 60 → $4800/100 =$ 48 km/h
3. Train and pole	$L \div s$	150 m train at 54 km/h (15 m/s) → 10 s
4. Train and platform	$(L + P) \div s$	200 m train, 300 m platform, 72 km/h (20 m/s) → $500 \div 20 =$ 25 s
5. Opposite trains	Add speeds	120 m and 180 m at 50 and 40 km/h → 90 km/h = 25 m/s → $300 \div 25 =$ 12 s
6. Same direction	Subtract speeds	100 m and 150 m at 70 and 52 km/h → 18 km/h = 5 m/s → $250 \div 5 =$ 50 s
7. Boat and stream	Half sum / half difference	Downstream 15, upstream 9 km/h → boat 12 , stream 3 km/h
8. Round trip by boat	$D/(u+v) + D/(u-v)$	Boat 10, stream 2, 48 km each way → $48/12 + 48/8 =$ 10 h
9. Late / early	$S_1 S_2 / (S_2 - S_1) \times \Delta t$	At 4 km/h, 10 min late; at 5 km/h, 5 min early → $20 \times \frac{1}{4} \text{ h} =$ 5 km
10. Slower than usual	Time $\times$ inverse ratio	At $\frac{3}{4}$ of usual speed he is 20 min late → extra $\frac{1}{3}$ of usual time = 20 → usual 60 min

Traps

- Convert km/h to m/s **before** using lengths in metres.
- A man or a pole has no length; a platform, bridge or another train does.
- Average speed is never the simple average of two speeds over equal distances.

B · REASONING — Mathematical Operations, Missing Numbers

Symbol substitution: rewrite the whole expression with the real signs first, then apply **BODMAS** ($\div$ and $\times$ before $+$ and $-$). **Missing numbers:** test rows, then columns, then the centre: sum, product, squares, cubes, or a combination.

Common rules in figures	Check
Row/column sum or product equal	Add or multiply each row
$a^2 + b^2 = c$	3, 4 $\rightarrow$ 25
Centre = product of corners	2, 3, 4 $\rightarrow$ 24
$(a + b) \times c$, $(a - b)^2$	Try after simple sums fail

Patterns, with solved examples

Pattern	Example	Solution
Substitution	If $+$ means $\times$, $\times$ means $-$, $-$ means $\div$, $\div$ means $+$: $8 + 4 \times 6 - 2 \div 5 = ?$	$8 \times 4 - 6 \div 2 + 5 = 32 - 3 + 5 = \mathbf{34}$
Interchange	Interchange $+$ and $\times$: $5 \times 3 + 2 = ?$	$5 + 3 \times 2 = \mathbf{11}$
Which swap?	Which two signs must be swapped to make $10 + 2 \times 3 - 4 = 8$ true?	Swap $+$ and $-$: $10 - 2 \times 3 + 4 = \mathbf{8}$
Grid	Rows: (3, 4, 25), (5, 12, 169), (6, 8, ?)	$a^2 + b^2$: $36 + 64 = \mathbf{100}$
Triangle centre	Corners 2, 3, 4 $\rightarrow$ 24; 3, 4, 5 $\rightarrow$ 60; 1, 5, 6 $\rightarrow$?	Product $\rightarrow \mathbf{30}$
Operation rule	$2 * 3 = 13$, $3 * 4 = 25$, $4 * 5 = ?$	$a^2 + b^2 \rightarrow 16 + 25 = \mathbf{41}$

Speed rule

Write the converted expression on paper before calculating — mental substitution plus BODMAS is where errors creep in.

C · SCIENCE — Chemistry 2: Chemical Reactions; Acids, Bases & Salts

8 lessons. Reactions and acids/bases/salts were 30 of 210 science questions in the 2026 papers (R). Most questions are direct: 'which type of reaction', 'which acid is in X', 'formula of Y'. The tables are the exam.

Lesson 1 · Chemical equations and balancing (about 7 min)

Signs that a reaction happened: a **gas** is given off, a **precipitate** forms, the **colour** or **temperature** changes, or the **state** changes. A balanced equation has the same number of each atom on both sides (law of conservation of mass).

Worked example — balancing



Balance Fe: **3Fe** · balance O: **4H₂O** · then H: **4H₂**



State symbols: (s) solid, (l) liquid, (g) gas, (aq) dissolved in water. ↑ = gas given off, ↓ = precipitate.

Quick check 1 — answer out loud before moving on (answers at the end)

Balance: $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}$

Lesson 2 · Types of chemical reactions (about 10 min)

Type	Pattern	Example
Combination	$A + B \rightarrow AB$	$\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$ (quicklime → slaked lime; lots of heat). Whitewash: $\text{Ca(OH)}_2 + \text{CO}_2 \rightarrow \text{CaCO}_3$ — the shiny finish
Decomposition	$AB \rightarrow A + B$	Heat: $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$ · electricity: $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$ ($\text{H}_2 : \text{O}_2 = 2 : 1$ by volume) · light: $2\text{AgCl} \rightarrow 2\text{Ag} + \text{Cl}_2$ (turns grey; B&W photography)
Displacement	$A + BC \rightarrow AC + B$	$\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$ (blue → light green; reddish copper coats the nail)
Double displacement	$AB + CD \rightarrow AD + CB$	$\text{Na}_2\text{SO}_4 + \text{BaCl}_2 \rightarrow \text{BaSO}_4\downarrow$ (white precipitate) + 2NaCl
Neutralisation	Acid + base → salt + water	$\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$

Heating green ferrous sulphate crystals: they turn white, then brown (Fe_2O_3), with a smell of burning sulphur — thermal decomposition.

Exam trap

Silver chloride is stored in **dark bottles** because light decomposes it. Most decomposition reactions need energy (heat, light or electricity).

Quick check 2 — answer out loud before moving on (answers at the end)

a) Type of reaction: $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$? b) Type: $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$

Lesson 3 · Exothermic/endothermic; oxidation and reduction (about 7 min)

	Meaning	Examples
Exothermic	Releases heat	Burning, respiration, quicklime + water, neutralisation
Endothermic	Absorbs heat	Photosynthesis, decomposition of CaCO_3 , dissolving NH_4Cl in water

Oxidation = gain of oxygen or loss of hydrogen. **Reduction** = loss of oxygen or gain of hydrogen. When both happen together it is a **redox** reaction. The substance that gives oxygen is the **oxidising agent**; the one that takes it is the **reducing agent**.

Worked example — $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$

CuO loses oxygen → **CuO is reduced**

H_2 gains oxygen → **H_2 is oxidised**

So H_2 is the reducing agent and CuO is the oxidising agent

Quick check 3 — answer out loud before moving on (answers at the end)

In $\text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}$, which substance is oxidised and which is reduced?

Lesson 4 · Corrosion and rancidity (about 6 min)

Rusting of iron needs **both air (oxygen) and water**. Rust is hydrated iron(III) oxide, **$\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$** . Prevention: paint, oil or grease; **galvanisation** (coating with zinc); alloying (stainless steel); electroplating (chromium, tin).

Other corrosion: copper gets a **green** coat (basic copper carbonate); silver turns **black** (silver sulphide).

Rancidity: oils and fats in food get **oxidised** and smell and taste bad. Prevention: antioxidants, airtight containers, refrigeration, and **flushing chip packets with nitrogen**.

Quick check 4 — answer out loud before moving on (answers at the end)

a) Coating iron with zinc is called? b) Why are chip packets filled with nitrogen?

Lesson 5 · Acids and bases: properties and indicators (about 9 min)

	Acids	Bases
Taste / feel	Sour	Bitter, soapy
In water give	H^+ (H_3O^+) ions	OH^- ions (water-soluble bases are alkalis : NaOH , KOH , $\text{Ca}(\text{OH})_2$, NH_4OH)
Litmus	Blue → red	Red → blue
Phenolphthalein	Colourless	Pink
Methyl orange	Red	Yellow
Turmeric	No change	Reddish-brown

Olfactory indicators (smell): **onion, vanilla, clove oil** lose their smell in a base. Litmus is extracted from **lichens**.

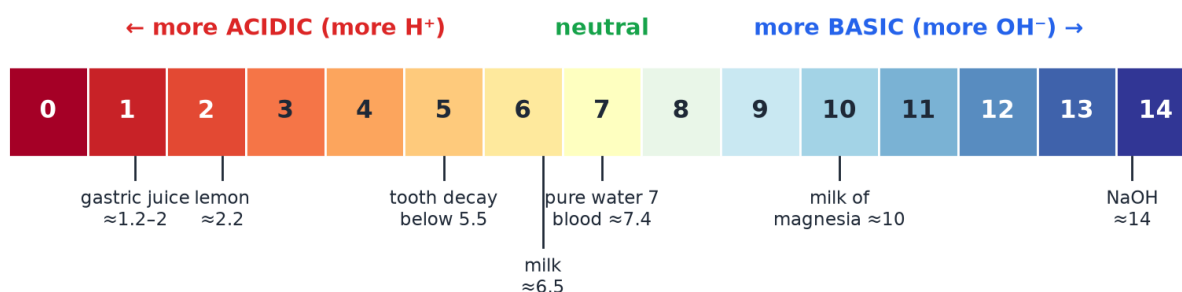
Source	Acid	Source	Acid
Vinegar	Acetic (ethanoic)	Ant sting, nettle	Methanoic (formic)
Lemon, orange	Citric	Curd, sour milk	Lactic
Tamarind, grapes	Tartaric	Tomato, spinach	Oxalic
Apple	Malic	Stomach	Hydrochloric

Reactions: acid + metal → salt + **hydrogen** (burns with a 'pop') · acid + carbonate → salt + water + **CO₂** (turns lime water milky) · acid + base → salt + water. **Always add acid to water**, never water to acid — the heat can splash it out.

Quick check 5 — answer out loud before moving on (answers at the end)

a) Acid in tomato? b) Colour of phenolphthalein in a base? c) Gas given off when zinc reacts with dilute HCl?

Lesson 6 · The pH scale (about 7 min)



pH runs from 0 to 14: **7 is neutral**, below 7 acidic, above 7 basic. A lower pH means more H⁺ ions. Universal indicator shows the whole range by colour.

Situation	pH fact	Remedy / meaning
Tooth decay	Starts when mouth pH falls below 5.5	Basic toothpaste neutralises the acid
Acidity / indigestion	Excess stomach HCl	Antacid: milk of magnesia Mg(OH) ₂ , baking soda
Bee sting	Acidic (methanoic acid)	Rub baking soda (a base)
Wasp sting	Alkaline	Apply vinegar (an acid)
Acid rain	pH below 5.6	From SO ₂ and NO _x
Human blood	≈ 7.35-7.45	Slightly basic

Quick check 6 — answer out loud before moving on (answers at the end)

a) Tooth decay starts below which pH? b) What is applied to a bee sting?

Lesson 7 · Important salts (about 9 min)

Common name	Chemical	Formula	Made / used
Common salt	Sodium chloride	NaCl	Raw material for NaOH, Cl ₂ , baking soda (chlor-alkali: brine → NaOH + Cl ₂ + H ₂)
Baking soda	Sodium hydrogen carbonate	NaHCO ₃	Baking powder (+ tartaric acid), antacid, soda-acid fire extinguisher
Washing soda	Sodium carbonate decahydrate	Na ₂ CO ₃ ·10H ₂ O	Removes permanent hardness; glass, soap, paper
Bleaching powder	Calcium oxychloride	CaOCl ₂	Ca(OH) ₂ + Cl ₂ ; disinfects drinking water, bleaches cotton
Plaster of Paris	Calcium sulphate hemihydrate	CaSO ₄ ·½H ₂ O	Gypsum heated at 373 K; sets with water into gypsum; casts for fractures, toys
Gypsum	Calcium sulphate dihydrate	CaSO ₄ ·2H ₂ O	Added to cement to slow setting
Blue vitriol	Copper sulphate pentahydrate	CuSO ₄ ·5H ₂ O	Heated → white anhydrous CuSO ₄ ; add water → blue again

Water of crystallisation = the fixed number of water molecules in a crystal: CuSO₄·**5**H₂O, CaSO₄·**2**H₂O, Na₂CO₃·**10**H₂O.

Quick check 7 — answer out loud before moving on (answers at the end)

a) Formula of plaster of Paris? b) Which salt removes permanent hardness of water?

Lesson 8 · Strong and weak acids and bases (about 5 min)

	Strong (fully ionised)	Weak (partly ionised)
Acids	HCl, H ₂ SO ₄ ('king of chemicals', 'oil of vitriol'), HNO ₃	Acetic, citric, carbonic, formic
Bases	NaOH (caustic soda), KOH (caustic potash)	NH ₄ OH, Mg(OH) ₂

Aqua regia = conc. HCl and conc. HNO₃ in the ratio **3 : 1**; it dissolves **gold and platinum**. Concentrated vs dilute is about the **amount** of acid in water; strong vs weak is about **how much it ionises**.

Quick check 8 — answer out loud before moving on (answers at the end)

What is the ratio of HCl to HNO₃ in aqua regia?

Science 1-page summary — revise from this tomorrow

Topic	Must remember
Reaction types	Combination $\text{CaO} + \text{H}_2\text{O}$ · decomposition CaCO_3 (heat), H_2O (electricity, $\text{H}_2 : \text{O}_2 = 2 : 1$), AgCl (light) · displacement $\text{Fe} + \text{CuSO}_4$ · double displacement $\text{BaSO}_4 \downarrow$
Redox	Oxidation = + O or – H · reduction = – O or + H · respiration exothermic, photosynthesis endothermic
Corrosion	Rust $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ needs air + water · galvanising = zinc · copper green, silver black · rancidity → N_2 in chip packets
Indicators	Litmus acid red / base blue · phenolphthalein base pink · methyl orange acid red / base yellow · turmeric base reddish-brown
Natural acids	Vinegar acetic · lemon citric · tamarind tartaric · tomato oxalic · curd lactic · ant sting formic · stomach HCl
pH	7 neutral · tooth decay < 5.5 · acid rain < 5.6 · blood ≈ 7.4 · bee sting → baking soda, wasp sting → vinegar
Salts	NaHCO_3 baking · $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ washing · CaOCl_2 bleaching · $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$ PoP · $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ gypsum · $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ blue vitriol
Strong / weak	HCl , H_2SO_4 , HNO_3 strong · aqua regia 3 HCl : 1 HNO_3 dissolves gold

Science practice set — 20 questions · target 6 minutes

#	Question
1	$\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$ is a: (a) decomposition reaction (b) combination reaction (c) displacement reaction (d) double displacement reaction
2	In the electrolysis of water, the volume ratio of H_2 to O_2 is: (a) 2 : 1 (b) 1 : 2 (c) 1 : 1 (d) 1 : 8
3	Silver chloride turns grey in sunlight because it: (a) absorbs moisture (b) is oxidised by air (c) melts (d) decomposes into silver and chlorine
4	When an iron nail is placed in copper sulphate solution, the solution turns: (a) deep blue (b) colourless (c) red (d) light green
5	$\text{Na}_2\text{SO}_4 + \text{BaCl}_2 \rightarrow \text{BaSO}_4 + 2\text{NaCl}$ is a: (a) combination reaction (b) decomposition reaction (c) redox reaction (d) double displacement reaction
6	Which is an endothermic process? (a) Respiration (b) Photosynthesis (c) Burning of coal (d) Quicklime + water
7	In $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$, the substance oxidised is: (a) CuO (b) H_2 (c) Cu (d) H_2O
8	Rusting of iron needs: (a) air only (b) water only (c) sunlight (d) both air and water
9	Galvanisation is coating iron with: (a) tin (b) zinc (c) chromium (d) copper
10	Chip packets are flushed with nitrogen to prevent: (a) rancidity (b) corrosion (c) evaporation (d) fermentation
11	Phenolphthalein in a basic solution is: (a) colourless (b) blue (c) pink (d) yellow
12	The acid in tamarind is: (a) citric acid (b) oxalic acid (c) tartaric acid (d) acetic acid
13	The acid in an ant sting is: (a) acetic acid (b) methanoic (formic) acid (c) lactic acid (d) citric acid
14	Acid + metal carbonate gives off: (a) hydrogen (b) oxygen (c) chlorine (d) carbon dioxide
15	Tooth decay starts when the pH in the mouth falls below: (a) 7.0 (b) 6.5 (c) 4.0 (d) 5.5
16	A solution with pH 11 is: (a) acidic (b) neutral (c) basic (d) strongly acidic
17	Chemical formula of baking soda: (a) NaHCO_3 (b) Na_2CO_3 (c) NaOH (d) CaCO_3
18	Plaster of Paris is: (a) $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$ (b) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ (c) CaCO_3 (d) CaOCl_2
19	Number of water molecules in a crystal of washing soda: (a) 5 (b) 2 (c) 10 (d) 7
20	Aqua regia is a mixture of conc. HCl and conc. HNO_3 in the ratio: (a) 1 : 3 (b) 3 : 1 (c) 1 : 1 (d) 2 : 1

Score	Time taken	Wrong Q numbers	Mistake type (concept / formula / calc / careless / recall)
/ 20			

Answers

Science practice set

#	Ans	Why / lesson to re-read
1	b	Lesson 2
2	a	Lesson 2
3	d	Lesson 2
4	d	Lesson 2
5	d	Lesson 2
6	b	Lesson 3
7	b	Lesson 3
8	d	Lesson 4
9	b	Lesson 4
10	a	Lesson 4
11	c	Lesson 5
12	c	Lesson 5
13	b	Lesson 5
14	d	Lesson 5
15	d	Lesson 6
16	c	Lesson 6
17	a	Lesson 7
18	a	Lesson 7
19	c	Lesson 7
20	b	Lesson 8

Quick checks

QC	Answers
1	$2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$
2	a) Decomposition (thermal) b) Displacement
3	Carbon (C) is oxidised to CO; zinc oxide (ZnO) is reduced to Zn
4	a) Galvanisation b) Nitrogen is unreactive and keeps oxygen out, so the oil in the chips doesn't turn rancid
5	a) Oxalic acid b) Pink c) Hydrogen
6	a) 5.5 b) Baking soda (a mild base)
7	a) $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$ b) Washing soda ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$)
8	3 : 1

Day 9 recall (Block 0)

#	Answer	#	Answer
1	Camphor, naphthalene, iodine, NH ₄ Cl, dry ice (any two)	6	3
2	Centrifugation	7	Ca(OH) ₂
3	Colloid	8	100
4	James Chadwick	9	6.022×10^{23}
5	18	10	6 days

Send me tonight

Day 9 recall score /10 · science practice score /20 and time · Maths and Reasoning: attempted, correct, average seconds per question · the question numbers you got wrong.